

System of Linear Equations

$AX=B$ $\longrightarrow$ How to determine the nature of solution?

$\downarrow$
why it is working?

Vector space (set + Binary operation)

Set $\rightarrow$ collection of well defined objects $\rightarrow$ Elements of the set.

function $\rightarrow$ A, B be two sets
 $f: A \rightarrow B$ is called a function

if $\forall a \in A \exists$ unique $b \in B$
s.t. $f(a) = b$.

Now Example:

A = set of all persons sitting in this room

B = set of their mobile number

$f: A \rightarrow B$

$f(\text{person}) = \text{his/her mobile number.}$

$f: B \rightarrow A$

$f(\text{mobile number}) = \text{its owner}$

Binary operation :- Let A be a set.

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A function from $A \times A$ to A

is called Binary operation.

$\mathbb{N} \rightarrow$ set of Natural number

$$= \{1, 2, \dots\}$$

$$\begin{aligned} \mathbb{N} \times \mathbb{N} &\rightarrow \mathbb{N} \\ (a, b) &\rightarrow a+b \\ (a, b) &\rightarrow ab \end{aligned}$$

$$\mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R}$$

$$(a, b) \rightarrow a/b$$

Not Binary operation

Not Binary operation

$$\begin{aligned} (a, b) &\rightarrow a/b \\ (2, 3) &\rightarrow 2/3 \notin \mathbb{N} \end{aligned}$$

$$\begin{aligned} \mathbb{Z} \times \mathbb{Z} &\rightarrow \mathbb{Z} \\ (a, b) &= a+b \end{aligned}$$

Vector space

Let $F = \mathbb{R}$ or $\mathbb{C}$. (field)

A non-empty set V is called vector space over F with binary operation $(+)$, addition and a scalar multiplication $(\cdot)$.

$$(F \times V \rightarrow V) \\ (\alpha, v) \rightarrow \alpha v, \quad \alpha \in F, v \in V,$$

satisfying the following axioms.

- ① $(v+w)+z = v+(w+z)$ (Associativity) ✓
- ② $\exists 0 \in V$ s.t. $v+0 = v = 0+v$ (Existence of zero)
- ③ $\forall v \in V, \exists -v \in V$ s.t. $v+(-v) = 0 = (-v)+v$. (Existence of Inverse)
- ④ $(v+w) = (w+v)$ (Commutativity) $\underline{v, w, z \in V}$
- ⑤ $(\alpha + \beta)v = \alpha v + \beta v$ $\alpha, \beta \in F, v \in V$
 $\alpha(v+w) = \alpha v + \alpha w$

$$\begin{aligned}
 (5) \quad & (\alpha + \beta)v = \alpha v + \beta v \quad \alpha, \beta \in \mathbb{F}, v \in V \\
 (6) \quad & (\alpha\beta)v = \alpha(\beta v) \quad (7) \quad \alpha(v+w) = \alpha v + \alpha w \\
 (8) \quad & 1 \cdot v = v
 \end{aligned}$$

Claim $\mathbb{R}^n$ forms v.s over $\mathbb{R}$

$$\mathbb{R}^n = \left\{ (x_1, x_2, \dots, x_n) \mid x_i \in \mathbb{R} \right\}$$

$$\begin{aligned}
 (x_1, x_2, \dots, x_n) + (y_1, y_2, \dots, y_n) &= (x_1 + y_1, x_2 + y_2, \dots, x_n + y_n) \\
 \alpha(x_1, x_2, \dots, x_n) &= (\alpha x_1, \alpha x_2, \dots, \alpha x_n)
 \end{aligned}$$

$$\begin{aligned}
 & ((x_1, x_2) + (y_1, y_2)) + (z_1, z_2) \\
 &= (x_1 + y_1, x_2 + y_2) + (z_1, z_2) \\
 &\stackrel{y_1 + x_1}{=} ((x_1 + y_1) + z_1, (x_2 + y_2) + z_2)
 \end{aligned}$$

$$(0, \dots, 0) + (x_1, \dots, x_n) = (x_1, \dots, x_n)$$

Inverse

$$\begin{aligned}
 & (x_1, \dots, x_n) \\
 & (-x_1, \dots, -x_n)
 \end{aligned}$$

$$\begin{aligned}
 & (\alpha + \beta)(x_1, x_2, \dots, x_n) \\
 &= \alpha(x_1, x_2, \dots, x_n) + \beta(x_1, x_2, \dots, x_n)
 \end{aligned}$$

$$\begin{aligned}
 & \alpha(\beta(x_1, x_2, \dots, x_n)) \\
 &= \alpha(\beta v)
 \end{aligned}$$

$$\alpha(v + w) = \alpha v + \alpha w$$

$$\alpha x_1 + \alpha y_1 = \alpha(x_1 + y_1)$$

* $\alpha(v+w) = \alpha v + \alpha w$

$\alpha v = \alpha(x_1, x_2, \dots, x_n)$ + $\alpha y_1 + \alpha y_2 = \alpha(y_1 + y_2)$

$\alpha w = \alpha(y_1, y_2, \dots, y_n)$

$\alpha(v+w) = \alpha(x_1+y_1, x_2+y_2, \dots, x_n+y_n)$

* $1 \cdot v = v$ $1 \cdot x_i = x_i = v$

* $\mathbb{R}^n$ is v.s over $\mathbb{C}$

$M_n(\mathbb{R}) = \left\{ A \mid A \text{ is } n \times n \text{ matrices over } \mathbb{R} \right\}$ ✓

$\begin{matrix} \text{is} \\ \mathbb{R}^{n^2} \end{matrix}$

$A+B$ (matrix addition)

αA

$M_n(\mathbb{R})$ is vector space over $\mathbb{R}$.

$M_2(\mathbb{R}) = \left\{ \begin{bmatrix} a & b \\ c & d \end{bmatrix} \mid a, b, c, d \in \mathbb{R} \right\}$

$\searrow$ $\mathbb{R}^4$ $(a, b, c, d) + (a', b', c', d')$

Exercise

$P_n(\mathbb{R}) = \{ a_0 + a_1x + \dots + a_nx^n \mid a_0, a_1, \dots, a_n \in \mathbb{R} \}$

Show that $P_n(\mathbb{R})$ forms vector space over $\mathbb{R}$ with polynomial addition and scalar multiplication.

$D_2(\mathbb{R}) = \left\{ \begin{bmatrix} a & 0 \\ 0 & b \end{bmatrix} \mid a, b \in \mathbb{R} \right\} \rightarrow \text{whether this is v.s over } \mathbb{R}??$

$$D_2(\mathbb{R}) = \left\{ \begin{bmatrix} a & 0 \\ 0 & b \end{bmatrix} \mid a, b \in \mathbb{R} \right\} \quad \text{v.s over } \mathbb{R}??$$

- ① Sum of two Diagonal matrices is Diagonal. $w_1 + w_2 \in W \Rightarrow w_1 + w_2 \in W$
- ② Scalar multiple of Diagonal matrix is Diagonal. $\alpha \in F, w \in W \Rightarrow \alpha w \in W$
- ③ Zero matrix is a Diagonal matrix. $0 \in W$
- ④ Negative (inverse) of diagonal matrix is a diagonal matrix. $w \in W, -w \in W$

vector space over $\mathbb{R}$ $\leftarrow D_2(\mathbb{R}) \subset M_2(\mathbb{R}) \rightarrow$ vector space over $\mathbb{R}$

$\rightarrow$ Subspace

Subspace: Let V be a vector space over F . A non-empty subset W of V is called Subspace of V if W itself is vector space over F .

- if
- ① $w_1, w_2 \in W \Rightarrow w_1 + w_2 \in W$
 - ② $w \in W, \alpha \in F \Rightarrow \alpha w \in W$
 - ③ $0 \in W$
 - ④ if $w \in W, -w \in W$

$$\textcircled{2} \Rightarrow \textcircled{4}$$

$$\alpha = -1$$

$$w \in W, (-1)w = -w \in W.$$

$$\textcircled{1}, \textcircled{4} \rightarrow$$

$$w \in W, -w \in W$$

$$\textcircled{1} \Rightarrow w + (-w) = 0 \in W.$$

so .

it

is the solution set

Example : Show that the solution set of the following system of linear equations is subspace of $\mathbb{R}^3$.

$$\begin{matrix} x+y+z=0 \\ x-y+z=0 \end{matrix} \rightarrow \boxed{\text{Infinitely many solutions.}}$$

- Solution set $\neq \emptyset$.
 $(0,0,0) \in W \rightarrow W \neq \emptyset$.

$$\begin{bmatrix} 1 & 1 & 1 \\ 1 & -1 & 1 \end{bmatrix} \xrightarrow{R_2 \rightarrow R_1 + R_2} \begin{bmatrix} 1 & 1 & 1 \\ 2 & 0 & 2 \end{bmatrix}$$

$$\left\{ (-z, 0, z) \mid z \in \mathbb{R} \right\}$$

$$\begin{matrix} x+y+z=0 \\ -2+y+z=0 \\ y=0 \end{matrix} \quad \begin{matrix} 2x=2z \\ x=z \end{matrix}$$

$$w_1, w_2 \in W, \quad w_1 + w_2 \in W$$

$$W = \left\{ (k, 0, -k) \mid k \in \mathbb{R} \right\}$$

$$(k_1, 0, -k_1) + (k_2, 0, -k_2) = (k_1 + k_2, 0, -(k_1 + k_2))$$

$$\alpha(k_1, 0, -k_1) = (\alpha k_1, 0, -\alpha k_1)$$

Subspace : Let V be a vector space over F .
 U is a subset of V called

Subspace: Let V be a vector space.
A non-empty subset W of V is called
subspace of V if

- ① $w_1, w_2 \in W \Rightarrow w_1 + w_2 \in W$
- ② $w \in W \Rightarrow \alpha w \in W \quad \forall \alpha \in F.$